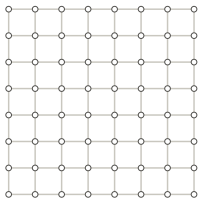
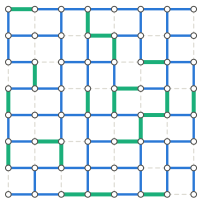
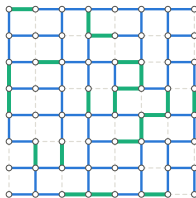


(a) Input

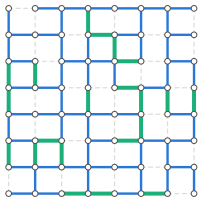
$n = 64, m = 112$
input graph

(b) Greedy

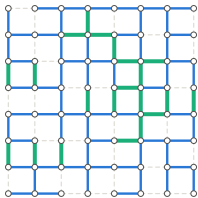
$D=3, C_E=2$
 $C_V=3, \Phi=48$

(c) Heap

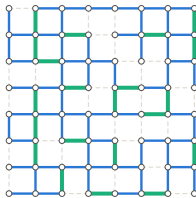
$D=3, C_E=2$
 $C_V=4, \Phi=60$

(d) Batch

$D=5, C_E=3$
 $C_V=3, \Phi=96$

(e) Fast

$D=5, C_E=3$
 $C_V=4, \Phi=120$

(f) Sample

$D=7, C_E=5$
 $C_V=6, \Phi=336$

Scaffold variants**Scaffold-Greedy**

Recompute scores after each edge insertion.

Scaffold-Heap

Cache scores; refresh affected candidates.

Scaffold-Batch

Score candidate batches; insert their top edges.

Scaffold-Fast

Score once; retain the highest-ranked edges.

Scaffold-Sample

Aggregate backbone scores; sample additional edges.

$\delta = 0.75, q = 84$

— RandSF backbone (63)

— Added edges (21)

--- Omitted edges (28)